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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 22th July Evening Shift

JEE Main Question Paper with Answer Key

22 Jul 2021	90	17	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	23 questions
Physics	27 questions
Out of syllabus	17 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

Which one of the following statements for D.I. Mendeleeff, is incorrect?

(A) He authored the textbook - Principles of Chemistry

(B) At the time, he proposed Periodic Table of elements structure of atom was known.

(C) Element with atomic number 101 is named after him.

(D) He invented accurate barometer.

Open solution

Q2. MCQ (Single Correct Answer)

Which purification technique is used for high boiling organic liquid compound (decomposes near its boiling point)?

(A) Simple distillation

(B) Steam distillation

(C) Fractional distillation

(D) Reduced pressure distillation

Open solution

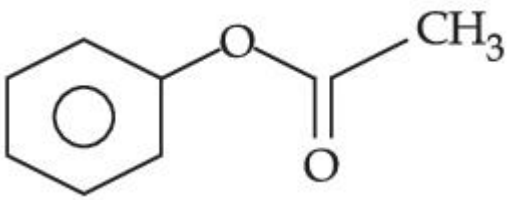
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Q3. MCQ (Single Correct Answer)

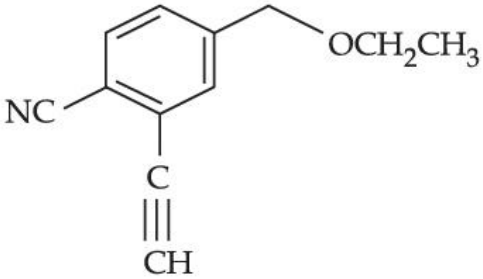
Open solution

Which of the following compounds will provide a tertiary alcohol on reaction with excess of CH_3MgBr followed by hydrolysis?

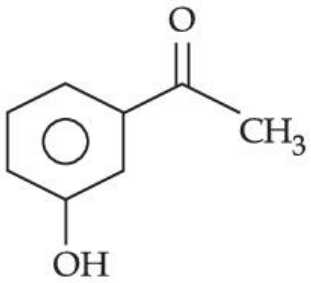
(A)



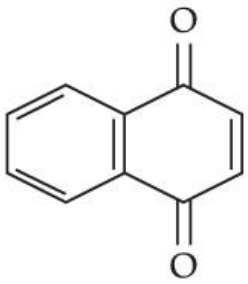
(B)



(C)



(D)



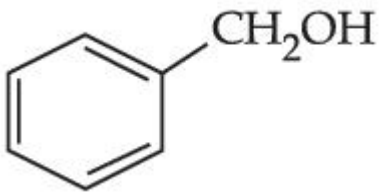
Q4. MCQ (Single Correct Answer)

Open solution

Which of the following compounds does not exhibit resonance?

(A) $CH_3CH_2OCH = CH_2$

(B)

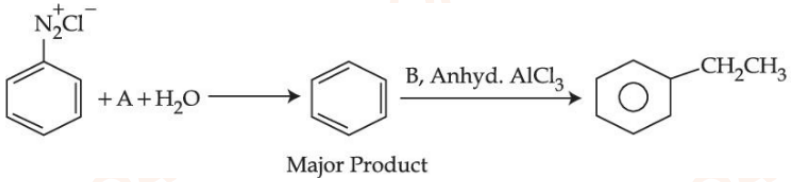


(C) $CH_3CH_2CH_2CONH_2$

(D) $CH_3CH_2CH = CHCH_2NH_2$

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Q5. MCQ (Single Correct Answer)



Major Product

Open solution

In the chemical reactions given above A and B respectively are :

(A) H_3PO_2 and CH_3CH_2Cl

(B) CH_3CH_2OH and H_3PO_2

(C) H_3PO_2 and CH_3CH_2OH

(D) CH_3CH_2Cl and H_3PO_2

Q6. MCQ (Single Correct Answer)

Open solution

Isotope(s) of hydrogen which emits low energy β^- particles with $t_{1/2}$ value > 12 years is/are

(A) Protium

(B) Tritium

(C) Deuterium

(D) Deuterium and Tritium

Q7. MCQ (Single Correct Answer)

Open solution

Match List-I with List-II :

	List-I (Species)		List-II (Hybrid Orbitals)
(a)	SF_4	(i)	sp^3d^2
(b)	IF_5	(ii)	d^2sp^3
(c)	NO_2^+	(iii)	sp^3d
(d)	NH_4^+	(iv)	sp^3
		(v)	sp

Choose the correct answer from the options given below :

(A) (a)-(i), (b)-(ii), (c)-(v) and (d)-(iii)

(B) (a)-(ii), (b)-(i), (c)-(iv) and (d)-(v)

(C) (a)-(iii), (b)-(i), (c)-(v) and (d)-(iv)

(D) (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(v)

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Q8. MCQ (Single Correct Answer)

Which of the following molecules does not show stereoisomerism?

(A) 3, 4-Dimethylhex-3-ene
(B) 3-Methylhex-1-ene
(C) 3-Ethylhex-3-ene
(D) 4-Methylhex-1-ene

Open solution

Q9. MCQ (Single Correct Answer)

Which one of the following group-15 hydride is the strongest reducing agent?

(A) AsH_3
(B) BiH_3
(C) PH_3
(D) SbH_3

Open solution

Q10. MCQ (Single Correct Answer)

The set having ions which are coloured and paramagnetic both is :

(A) Cu^{2+}, Cr^{3+}, Sc^{+}
(B) Cu^{2+}, Zn^{2+}, Mn^{4+}
(C) Sc^{3+}, V^{5+}, Ti^{4+}
(D) Ni^{2+}, Mn^{7+}, Hg^{2+}

Open solution

Q11. MCQ (Single Correct Answer)

Thiamine and pyridoxine are also known respectively as :

(A) Vitamin B_2 and Vitamin E
(B) Vitamin E and Vitamin B_2
(C) Vitamin B_6 and Vitamin B_2
(D) Vitamin B_1 and Vitamin B_6

Open solution

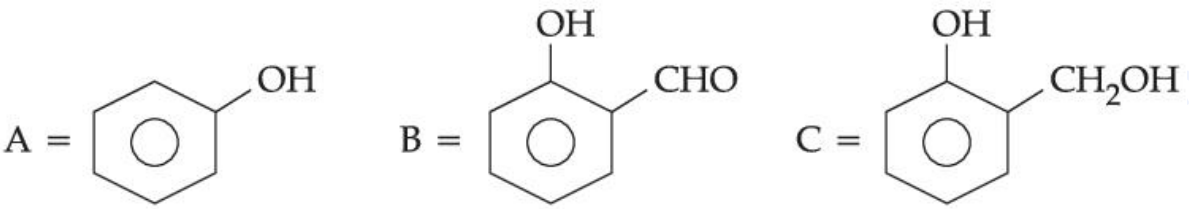
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Q12. MCQ (Single Correct Answer)

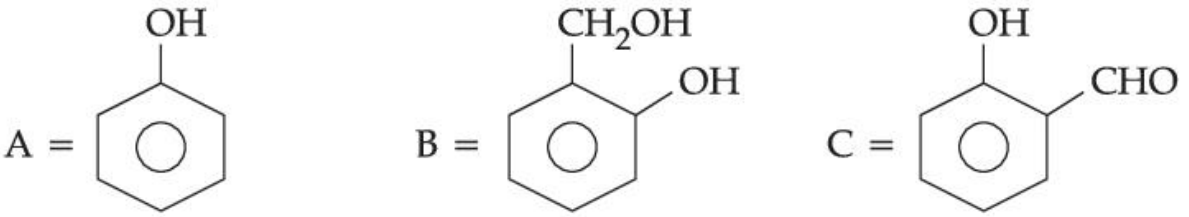
Open solution

An organic compound A (C_6H_6O) gives dark green colouration with ferric chloride. On treatment with $CHCl_3$ and KOH, followed by acidification gives compound B. Compound B can also be obtained from compound C on reaction with pyridinium chlorochromate (PCC). Identify A, B and C.

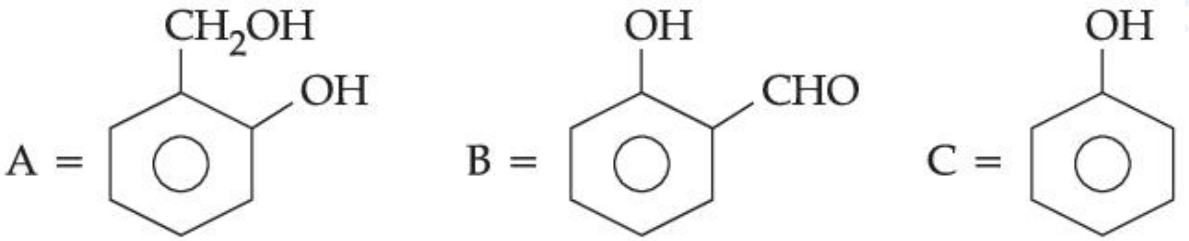
(A)



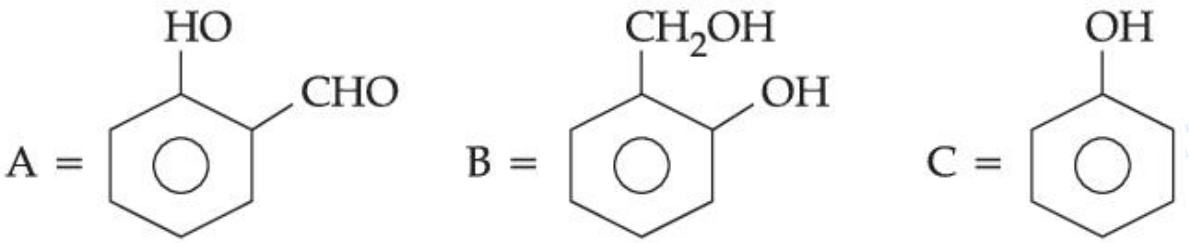
(B)



(C)



(D)



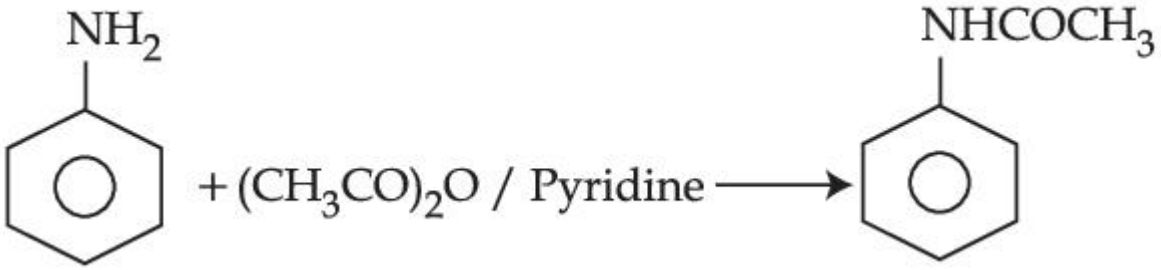
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Q13. MCQ (Single Correct Answer)

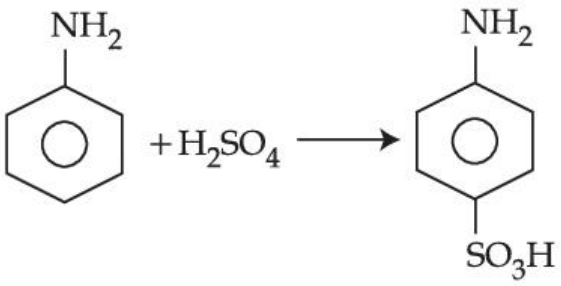
Open solution

Which one of the following reactions does not occur?

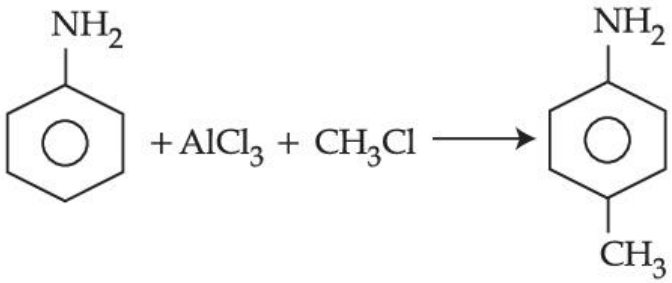
(A)



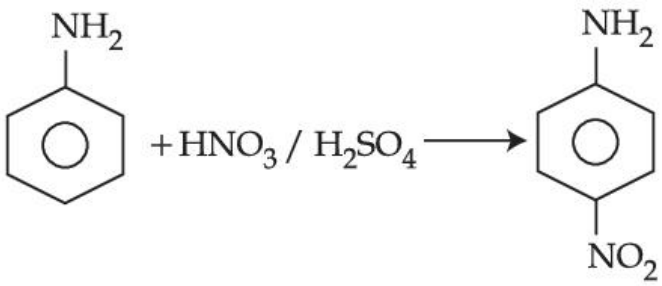
(B)



(C)



(D)



Q14. MCQ (Single Correct Answer)

Open solution

Which one of the following 0.06 M aqueous solutions has lowest freezing point?

(A) $Al_2(SO_4)_3$ (B) $C_6H_{12}O_6$ (C) KI (D) K_2SO_4

Q15. Numerical

Open solution

The total number of unpaired electrons present in $[Co(NH_3)_6]Cl_2$ and $[Co(NH_3)_6]Cl_3$ is :

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Q16. Numerical

Methylation of 10 g of benzene gave 9.2 g of toluene. Calculate the percentage yield of toluene _____. (Nearest integer)

Open solution

Q17. Numerical

The number of acyclic structural isomers (including geometrical isomers) for pentene are _____

Open solution

Q18. Numerical

Assume a cell with the following reaction

$Cu_{(s)} + 2Ag^{+}(1 \times 10^{-3}M) \rightarrow Cu^{2+}(0.250M) + 2Ag_{(s)}$

$E^{\Theta}_{cell} = 2.97$ V

E_{cell} for the above reaction is _____ V. (Nearest integer)

[Given : log 2.5 = 0.3979, T = 298 K]

Open solution

Q19. Numerical

Value of K_P for the equilibrium reaction

$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ at 288 K is 47.9. The K_C for this reaction at same temperature is _____. (Nearest integer)

(R = 0.083 L bar K^{-1} mol^{-1})

Open solution

Q20. Numerical

If the standard molar enthalpy change for combustion of graphite powder is -2.48×10^2 kJ mol^{-1}, the amount of heat generated on combustion of 1 g of graphite powder is _____ kJ. (Nearest integer)

Open solution

Q21. Numerical

Number of electrons that Vanadium (Z = 23) has in p-orbitals is equal to _____

Open solution

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Q22. Numerical

$N_2O_{5(g)} \rightarrow 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$

In the above first order reaction the initial concentration of N_2O_5 is 2.40×10^{-2} mol L^{-1} at 318 K. The concentration of N_2O_5 after 1 hour was 1.60×10^{-2} mol L^{-1}. The rate constant of the reaction at 318 K is _____ $\times 10^{-3}$ min^{-1}. (Nearest integer)

[Given : log 3 = 0.477, log 5 = 0.699]

Open solution

Q23. Numerical

If the concentration of glucose ($C_6H_{12}O_6$) in blood is 0.72 g L^{-1}, the molarity of glucose in blood is _____ $\times 10^{-3}$ M. (Nearest integer)

[Given : Atomic mass of C = 12, H = 1, O = 16 u]

Open solution

Mathematics 23 questions

Q24. MCQ (Single Correct Answer)

Let S_n denote the sum of first n-terms of an arithmetic progression. If S_{10} = 530, S_5 = 140, then $S_{20} - S_6$ is equal to:

(A) 1862 (B) 1842 (C) 1852 (D) 1872

Open solution

Q25. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be defined as

$f(x) = \begin{cases} -\frac{4}{3}x^3 + 2x^2 + 3x, & x > 0 \\ 3xe^x, & x \leq 0 \end{cases}$. Then f is increasing function in the interval

(A) $\left(-\frac{1}{2}, 2\right)$ (B) (0,2)

(C) $\left(-1, \frac{3}{2}\right)$ (D) $(-3, -1)$

Open solution

Q26. MCQ (Single Correct Answer)

Let y = y(x) be the solution of the differential equation $\text{cosec}^2 x\, dy + 2dx = (1 + y\cos 2x)\text{cosec}^2 x\, dx$, with $y\left(\frac{\pi}{4}\right) = 0$. Then, the value of $(y(0) + 1)^2$ is equal to :

(A) $e^{1/2}$ (B) $e^{-1/2}$ (C) e^{-1} (D) e

Open solution

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Q27. MCQ (Single Correct Answer)

Four dice are thrown simultaneously and the numbers shown on these dice are recorded in 2×2 matrices. The probability that such formed matrix have all different entries and are non-singular, is :

(A) $\frac{45}{162}$ (B) $\frac{21}{81}$ (C) $\frac{22}{81}$ (D) $\frac{43}{162}$

Open solution

Q28. MCQ (Single Correct Answer)

If $\int_0^{100\pi} \frac{\sin^2 x}{e^{\left(\frac{x}{\pi}-\left[\frac{x}{\pi}\right]\right)}} dx = \frac{\alpha\pi^3}{1+4\pi^2}, \alpha \in R$ where [x] is the greatest integer less than or equal to x, then the value of α is :

(A) 200 $(1 - e^{-1})$ (B) 100 $(1 - e)$ (C) 50 $(e - 1)$ (D) 150 $(e^{-1} - 1)$

Open solution

Q29. MCQ (Single Correct Answer)

The values of λ and μ such that the system of equations $x + y + z = 6, 3x + 5y + 5z = 26$, $x + 2y + \lambda z = \mu$ has no solution, are :

(A) $\lambda = 3, \mu = 5$ (B) $\lambda = 3, \mu \neq 10$

(C) $\lambda \neq 2, \mu = 10$ (D) $\lambda = 2, \mu \neq 10$

Open solution

Q30. MCQ (Single Correct Answer)

If the shortest distance between the straight lines $3(x-1) = 6(y-2) = 2(z-1)$ and $4(x-2) = 2(y-\lambda) = (z-3), \lambda \in R$ is $\frac{1}{\sqrt{38}}$, then the integral value of λ is equal to :

(A) 3 (B) 2 (C) 5 (D) -1

Open solution

Q31. MCQ (Single Correct Answer)

Let A = $[a_{ij}]$ be a real matrix of order 3×3, such that $a_{i1} + a_{i2} + a_{i3} = 1$, for i = 1, 2, 3. Then, the sum of all the entries of the matrix A^3 is equal to :

(A) 2 (B) 1 (C) 3 (D) 9

Open solution

Q32. MCQ (Single Correct Answer)

Let [x] denote the greatest integer less than or equal to x. Then, the values of x$\in$R satisfying the equation $[e^x]^2 + [e^x + 1] - 3 = 0$ lie in the interval :

(A) $\left[0, \frac{1}{e}\right)$ (B) [$\log_e 2$, $\log_e 3$) (C) [1, e) (D) [0, $\log_e 2$)

Open solution

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Q33. MCQ (Single Correct Answer)

Let the circle S : $36x^2 + 36y^2 - 108x + 120y + C = 0$ be such that it neither intersects nor touches the co-ordinate axes. If the point of intersection of the lines, $x - 2y = 4$ and $2x - y = 5$ lies inside the circle S, then :

Open solution

(A) $\frac{25}{9} < C < \frac{13}{3}$ (B) $100 < C < 165$

(C) $81 < C < 156$ (D) $100 < C < 156$

Q34. MCQ (Single Correct Answer)

Let n denote the number of solutions of the equation $z^2 + 3\overline{z} = 0$, where z is a complex number. Then the value of $\sum_{k=0}^{\infty} \frac{1}{n^k}$ is equal to :

Open solution

(A) 1 (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) 2

Q35. MCQ (Single Correct Answer)

If the domain of the function $f(x) = \frac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\frac{2x-1}{2}\right)}}$ is the interval $(\alpha, \beta]$, then $\alpha + \beta$ is equal to :

Open solution

(A) $\frac{3}{2}$ (B) 2 (C) $\frac{1}{2}$ (D) 1

Q36. MCQ (Single Correct Answer)

Let f : R $\to$ R be defined as $f(x) = \begin{cases} \frac{x^3}{(1-\cos 2x)^2}\log_e\left(\frac{1+2xe^{-2x}}{(1-xe^{-x})^2}\right), & x \neq 0 \\ \alpha, & x = 0 \end{cases}$

Open solution

If f is continuous at x = 0, then α is equal to :

(A) 1 (B) 3 (C) 0 (D) 2

Q37. MCQ (Single Correct Answer)

Let $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$. Let E_2 be another ellipse such that it touches the end points of major axis of E_1 and the foci of E_2 are the end points of minor axis of E_1. If E_1 and E_2 have same eccentricities, then its value is :

Open solution

(A) $\frac{-1+\sqrt{5}}{2}$ (B) $\frac{-1+\sqrt{8}}{2}$ (C) $\frac{-1+\sqrt{3}}{2}$ (D) $\frac{-1+\sqrt{6}}{2}$

Q38. Numerical

Let A = {0, 1, 2, 3, 4, 5, 6, 7}. Then the number of bijective functions f : A $\to$ A such that f(1) + f(2) = 3 $-$ f(3) is equal to

Open solution

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Q39. Numerical

If the digits are not allowed to repeat in any number formed by using the digits 0, 2, 4, 6, 8, then the number of all numbers greater than 10,000 is equal to _____.

Open solution

Q40. Numerical

Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then the number of 3 $\times$ 3 matrices B with entries from the set {1, 2, 3, 4, 5} and satisfying AB = BA is _____.

Open solution

Q41. Numerical

Consider the following frequency distribution :

Class :	0-6	6-12	12-18	18-24	24-30
Frequency :	a	b	12	9	5

If mean = $\frac{309}{22}$ and median = 14, then the value $(a - b)^2$ is equal to _____.

Open solution

Q42. Numerical

The sum of all the elements in the set {n$\in$ {1, 2, ..., 100} | H.C.F. of n and 2040 is 1} is equal to _____.

Open solution

Q43. Numerical

The area (in sq. units) of the region bounded by the curves $x^2 + 2y - 1 = 0$, $y^2 + 4x - 4 = 0$ and $y^2 - 4x - 4 = 0$, in the upper half plane is _____.

Open solution

Q44. Numerical

Let f : R $\rightarrow$ R be a function defined as $f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & if \quad |x| \leq 2 \\ 0 & if \quad |x| > 2 \end{cases}$

Let g : R $\rightarrow$ R be given by $g(x) = f(x + 2) - f(x - 2)$. If n and m denote the number of points in R where g is not continuous and not differentiable, respectively, then n + m is equal to _____.

Open solution

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Q45. Numerical

If the constant term, in binomial expansion of $\left(2x^r + \frac{1}{x^2}\right)^{10}$ is 180, then r is equal to _____.

Open solution

Q46. Numerical

Let y = y(x) be the solution of the differential equation $\left((x+2)e^{\left(\frac{y+1}{x+2}\right)} + (y+1)\right)dx = (x+2)dy$, y(1) = 1. If the domain of y = y(x) is an open interval (α, β), then $|\alpha + \beta|$ is equal to _____.

Open solution

Physics 27 questions

Q47. MCQ (Single Correct Answer)

What will be the projection of vector $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ on vector $\vec{B} = \hat{i} + \hat{j}$?

(A) $\sqrt{2}(\hat{i} + \hat{j} + \hat{k})$ (B) $(\hat{i} + \hat{j})$

(C) $\sqrt{2}(\hat{i} + \hat{j})$ (D) $2(\hat{i} + \hat{j} + \hat{k})$

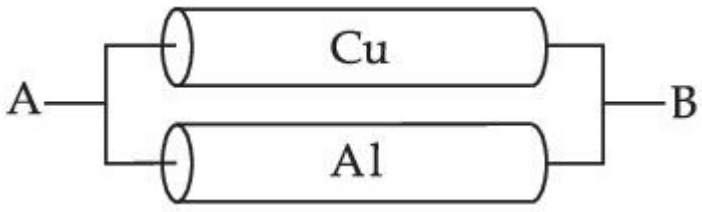
Open solution

Q48. MCQ (Single Correct Answer)

A Copper (Cu) rod of length 25 cm and cross-sectional area 3 mm^2 is joined with a similar Aluminium (Al) rod as shown in figure. Find the resistance of the combination between the ends A and B.

Open solution

(Take Resistivity of Copper = 1.7×10^{-8} Ωm and Resistivity of Aluminium = 2.6×10^{-8} Ωm)



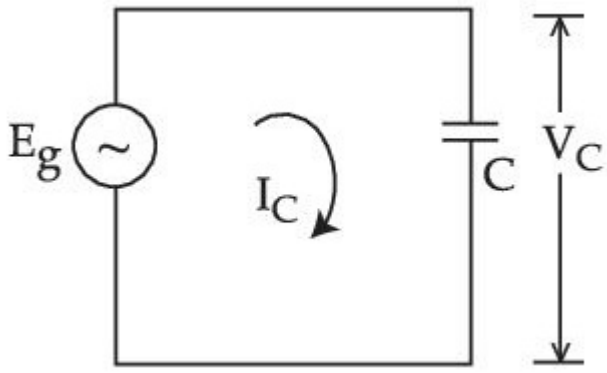
(A) 0.0858 mΩ (B) 1.420 mΩ (C) 0.858 mΩ (D) 2.170 mΩ

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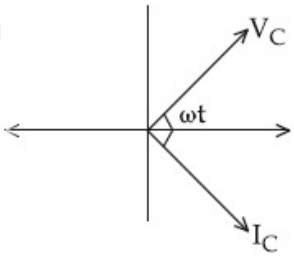
Q49. MCQ (Single Correct Answer)

In a circuit consisting of a capacitance and a generator with alternating emf $E_g = E_{g_0} \sin\omega t$, V_C and I_C are the voltage and current. Correct phasor diagram for such circuit is

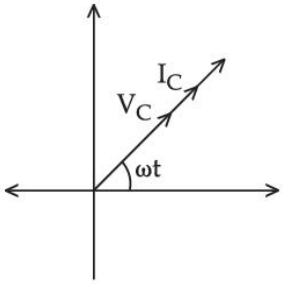
Open solution



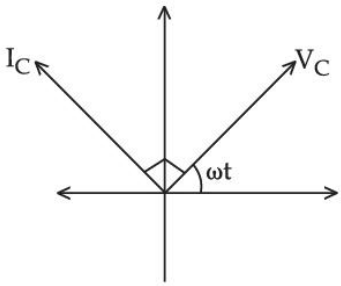
(A)



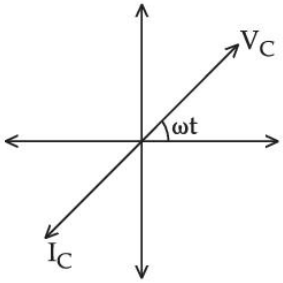
(B)



(C)



(D)



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Q50. MCQ (Single Correct Answer)

A porter lifts a heavy suitcase of mass 80 kg and at the destination lowers it down by a distance of 80 cm with a constant velocity. Calculate the work done by the porter in lowering the suitcase.

Open solution

(take g = 9.8 ms^{-2})

(A) +627.2 J (B) −62720.0 J (C) −627.2 J (D) 784.0 J

Q51. MCQ (Single Correct Answer)

Consider a situation in which a ring, a solid cylinder and a solid sphere roll down on the same inclined plane without slipping. Assume that they start rolling from rest and having identical diameter.

Open solution

The correct statement for this situation is

(A) All of them will have same velocity.

(B) The ring has greatest and the cylinder has the least velocity of the centre of mass at the bottom of the inclined plane.

(C) The sphere has the greatest and the ring has the least velocity of the centre of mass at the bottom of the inclined plane.

(D) The cylinder has the greatest and the sphere has the least velocity of the centre of mass at the bottom of the inclined plane.

Q52. MCQ (Single Correct Answer)

Intensity of sunlight is observed as 0.092 Wm^{-2} at a point in free space. What will be the peak value of magnetic field at the point?

Open solution

($\varepsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$)

(A) 2.77×10^{-8} T (B) 1.96×10^{-8} T (C) 8.31 T (D) 5.88 T

Q53. MCQ (Single Correct Answer)

A body is projected vertically upwards from the surface of earth with a velocity sufficient enough to carry it to infinity. The time taken by it to reach height h is _____ s.

Open solution

(A) $\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{\frac{3}{2}}-1\right]$

(B) $\frac{1}{3}\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{\frac{3}{2}}-1\right]$

(C) $\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{\frac{3}{2}}-1\right]$

(D) $\frac{1}{3}\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{\frac{3}{2}}-1\right]$

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Q54. MCQ (Single Correct Answer)

An electric dipole is placed on x-axis in proximity to a line charge of linear charge density 3.0×10^{-6} C/m. Line charge is placed on z-axis and positive and negative charge of dipole is at a distance of 10 mm and 12 mm from the origin respectively. If total force of 4N is exerted on the dipole, find out the amount of positive or negative charge of the dipole.

Open solution

(A) 0.485 mC (B) 815.1 nC (C) 8.8 μC (D) 4.44 μC

Q55. MCQ (Single Correct Answer)

A nucleus with mass number 184 initially at rest emits an α-particle. If the Q value of the reaction is 5.5 MeV, calculate the kinetic energy of the α-particle.

Open solution

(A) 5.5 MeV (B) 5.0 MeV (C) 5.38 MeV (D) 0.12 MeV

Q56. MCQ (Single Correct Answer)

A bullet of '4 g' mass is fired from a gun of mass 4 kg. If the bullet moves with the muzzle speed of 50 ms^{-1}, the impulse imparted to the gun and velocity of recoil of gun are :

Open solution

(A) 0.2 kg ms^{-1}, 0.1 ms^{-1} (B) 0.4 kg ms^{-1}, 0.05 ms^{-1}

(C) 0.2 kg ms^{-1}, 0.05 ms^{-1} (D) 0.4 kg ms^{-1}, 0.1 ms^{-1}

Q57. MCQ (Single Correct Answer)

Statement I : The ferromagnetic property depends on temperature. At high temperature, ferromagnet becomes paramagnet.

Open solution

Statement II : At high temperature, the domain wall area of a ferromagnetic substance increases.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

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Q58. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II

	List - I		List - II
(a)	$\omega L > \frac{1}{\omega C}$	(i)	Current is in phase with emf
(b)	$\omega L = \frac{1}{\omega C}$	(ii)	Current lags behind the applied emf
(c)	$\omega L < \frac{1}{\omega C}$	(iii)	Maximum current occurs
(d)	Resonant frequency	(iv)	Current leads the emf

Choose the correct answer from the options given below

(A) a(ii), b(i), c(iv), d(iii)

(B) a(ii), b(i), c(iii), d(iv)

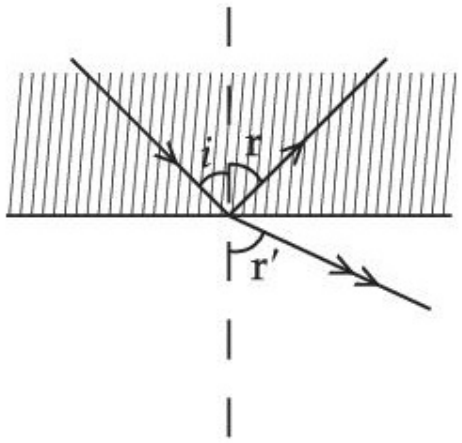
(C) a(iii), b(i), c(iv), d(ii)

(D) a(iv), b(iii), c(ii), d(i)

Q59. MCQ (Single Correct Answer)

Open solution

A ray of light passes from a denser medium to a rarer medium at an angle of incidence i. The reflected and refracted rays make an angle of 90° with each other. The angle of reflection and refraction are respectively r and r'. The critical angle is given by



(A) $\sin^{-1}$ (tan r)

(B) $\sin^{-1}$ (cot r)

(C) $\sin^{-1}$ (tan r')

(D) $\tan^{-1}$ (sin i)

Q60. MCQ (Single Correct Answer)

Open solution

An electron of mass m_e and a proton of mass m_p are accelerated through the same potential difference. The ratio of the de-Broglie wavelength associated with the electron to that with the proton is

(A) $\frac{m_e}{m_p}$

(B) 1

(C) $\frac{m_p}{m_e}$

(D) $\sqrt{\frac{m_p}{m_e}}$

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Q61. MCQ (Single Correct Answer)

What will be the average value of energy for a monoatomic gas in thermal equilibrium at temperature T?

(A) $\frac{3}{2}k_BT$ (B) k_BT (C) $\frac{2}{3}k_BT$ (D) $\frac{1}{2}k_BT$

Open solution

Q62. MCQ (Single Correct Answer)

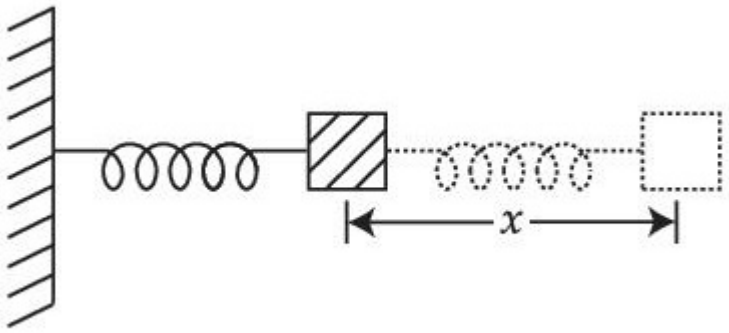
T_0 is the time period of a simple pendulum at a place. if the length of the pendulum is reduced to $\frac{1}{16}$ times of its initial value, the modified time period is :

(A) $4\,T_0$ (B) $\frac{1}{4}\,T_0$ (C) T_0 (D) $8\pi\,T_0$

Open solution

Q63. MCQ (Single Correct Answer)

The motion of a mass on a spring, with spring constant K is as shown in figure.



Open solution

The equation of motion is given by
x(t) = A sinωt + B cosωt with $\omega = \sqrt{\frac{K}{m}}$

Suppose that at time t = 0, the position of mass is x(0) and velocity v(0), then its displacement can also be represented as x(t) = C cos(ωt $-$ ϕ), where C and ϕ are :

(A) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{x(0)\omega}{2v(0)}\right)$

(B) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{x(0)\omega}{v(0)}\right)$

(C) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{v(0)}{x(0)\omega}\right)$

(D) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{v(0)}{x(0)\omega}\right)$

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Q64. MCQ (Single Correct Answer)

Consider a situation in which reverse biased current of a particular P-N junction increases when it is exposed to a light of wavelength $\leq$ 621 nm. During this process, enhancement in carrier concentration takes place due to generation of hole-electron pairs. The value of band gap is nearly.

(A) 1 eV (B) 4 eV (C) 0.5 eV (D) 2 eV

Open solution

Q65. Numerical

A ray of light passing through a prism ($\mu = \sqrt{3}$) suffers minimum deviation. It is found that the angle of incidence is double the angle of refraction within the prism. Then, the angle of prism is _____ (in degrees).

Open solution

Q66. Numerical

Three students S_1, S_2 and S_3 perform an experiment for determining the acceleration due to gravity (g) using a simple pendulum. They use different lengths of pendulum and record time for different number of oscillations. The observations are as shown in the table.

Student No.	Length of Pendulum (cm)	No. of oscillations (n)	Total time for n oscillations	Time period (s)
1	64.0	8	128.0	16.0
2	64.0	4	64.0	16.0
3	20.0	4	36.0	9.0

(Least count of length = 0.1 cm and Least count for time = 0.1 s)

If E_1, E_2 and E_3 are the percentage errors in 'g' for students 1, 2 and 3 respectively, then the minimum percentage error is obtained by student no. _____.

Open solution

Q67. Numerical

Three particles P, Q and R are moving along the vectors $\vec{A} = \hat{i} + \hat{j}$, $\vec{B} = \hat{j} + \hat{k}$ and $\vec{C} = -\hat{i} + \hat{j}$ respectively. They strike on a point and start to move in different directions. Now particle P is moving normal to the plane which contains vector $\vec{A}$ and $\vec{B}$. Similarly particle Q is moving normal to the plane which contains vector $\vec{A}$ and $\vec{C}$. The angle between the direction of motion of P and Q is $\cos^{-1}\left(\frac{1}{\sqrt{x}}\right)$. Then the value of x is _____.

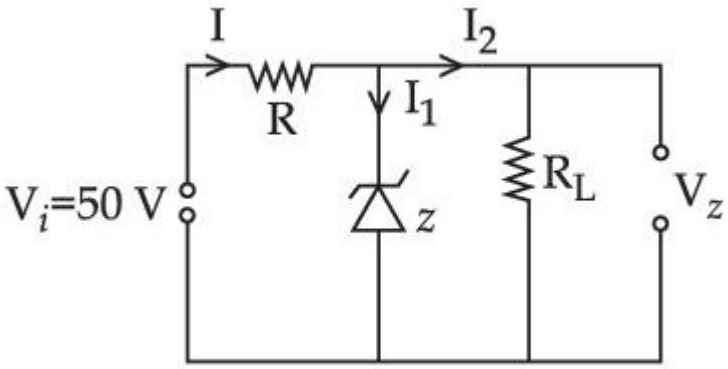
Open solution

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Q68. Numerical

In a given circuit diagram, a 5 V zener diode along with a series resistance is connected across a 50 V power supply. The minimum value of the resistance required, if the maximum zener current is 90 mA will be _____ Ω.

Open solution



Q69. Numerical

The total charge enclosed in an incremental volume of 2×10^{-9} m^3 located at the origin is _____ nC, if electric flux density of its field is found as

$D = e^{-x} \sin y\, \hat{i} - e^{-x} \cos y\, \hat{j} + 2z\, \hat{k}$ C/m^2

Open solution

Q70. Numerical

The position of the centre of mass of a uniform semi-circular wire of radius 'R' placed in x-y plane with its centre at the origin and the line joining its ends as x-axis is given by $\left(0, \frac{xR}{\pi}\right)$. Then, the value of | x | is _____.

Open solution

Q71. Numerical

In an electric circuit, a cell of certain emf provides a potential difference of 1.25 V across a load resistance of 5Ω. However, it provides a potential difference of 1 V across a load resistance of 2Ω. The emf of the cell is given by $\frac{x}{10}V$. Then the value of x is _____.

Open solution

Q72. Numerical

The area of cross-section of a railway track is 0.01 m^2. The temperature variation is 10°C. Coefficient of linear expansion of material of track is 10^{-5}/°C. The energy stored per meter in the track is _____ J/m.

(Young's modulus of material of track is 10^{11} Nm^{-2})

Open solution

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Q73. Numerical

The centre of a wheel rolling on a plane surface moves with a speed v_0. A particle on the rim of the wheel at the same level as the centre will be moving at a speed $\sqrt{x}v_0$. Then the value of x is _____.

Open solution

Out of syllabus 17 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q74. MCQ (Single Correct Answer) · **Chemistry**

The water having more dissolved O_2 is :

(A) boiling water (B) water at 80°C (C) polluted water (D) water at 4°C

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Match List-I with List-II

	List-I (Elements)		List-II (Properties)
(a)	Ba	(i)	Organic solvent soluble compounds
(b)	Ca	(ii)	Outer electronic configuration $6s^2$
(c)	Li	(iii)	Oxalate insoluble in water
(d)	Na	(iv)	Formation of very strong monoacidic base

Choose the correct answer from the options given below :

(A) (a)-(ii), (b)-(iii), (c)-(i) and (d)-(iv)

(B) (a)-(iv), (b)-(i), (c)-(ii) and (d)-(iii)

(C) (a)-(iii), (b)-(ii), (c)-(iv) and (d)-(i)

(D) (a)-(i), (b)-(iv), (c)-(ii) and (d)-(iii)

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

When silver nitrate solution is added to potassium iodide solution then the sol produced is :

(A) AgI / I^- (B) AgI / Ag^+ (C) KI / NO_3^- (D) $AgNO_3$ / NO_3^-

Open solution

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Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are the statements about diborane

(a) Diborane is prepared by the oxidation of $NaBH_4$ with I_2

(b) Each boron atom is in sp^2 hybridized state

(c) Diborane has one bridged 3 centre-2-electron bond

(d) Diborane is a planar molecule

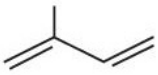
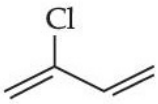
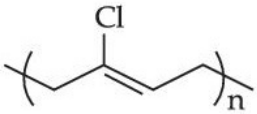
The option with correct statements is :

(A) (c) and (d) only **(B)** (a) only **(C)** (c) only **(D)** (a) and (b) only

Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List-I with List-II :

List - I		List - II	
(a)	Chloroprene	(i)	
(b)	Neoprene	(ii)	
(c)	Acrylonitrile	(iii)	
(d)	Isoprene	(iv)	$CH_2=CH-CN$

Choose the correct answer from the options given below :

(A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i) **(B)** (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii) **(D)** (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Sulphide ion is soft base and its ores are common for metals.

(a) Pb (b) Al (c) Ag (d) Mg

(A) (a) and (c) only **(B)** (a) and (d) only **(C)** (a) and (b) only **(D)** (c) and (d) only

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Q80. Numerical · **Chemistry**

A copper complex crystallising in a CCP lattice with a cell edge of 0.4518 nm has been revealed by employing X-ray diffraction studies. The density of a copper complex is found to be 7.62 g cm^{-3}. The molar mass of copper complex is _____ g mol^{-1}. (Nearest integer)

[Given : N_A = 6.022 $\times$ 10^{23} mol^{-1}]

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let L be the line of intersection of planes $\vec{r}.(\hat{i}-\hat{j}+2\hat{k})=2$ and $\vec{r}.(2\hat{i}+\hat{j}-\hat{k})=2$. If $P(\alpha,\beta,\gamma)$ is the foot of perpendicular on L from the point (1, 2, 0), then the value of $35(\alpha+\beta+\gamma)$ is equal to :

(A) 101 (B) 119 (C) 143 (D) 134

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let a vector $\vec{a}$ be coplanar with vectors $\vec{b}=2\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$. If $\vec{a}$ is perpendicular to $\vec{d}=3\hat{i}+2\hat{j}+6\hat{k}$, and $|\vec{a}|=\sqrt{10}$. Then a possible value of $[\vec{a}\ \vec{b}\ \vec{c}]+[\vec{a}\ \vec{b}\ \vec{d}]+[\vec{a}\ \vec{c}\ \vec{d}]$ is equal to :

(A) -42 (B) -40 (C) -29 (D) -38

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let three vectors $\vec{a},\vec{b}$ and $\vec{c}$ be such that $\vec{a}\times\vec{b}=\vec{c},\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{a}|=2$. Then which one of the following is not true?

(A) $\vec{a}\times\left((\vec{b}+\vec{c})\times(\vec{b}-\vec{c})\right)=\vec{0}$

(B) Projection of $\vec{a}$ on $(\vec{b}\times\vec{c})$ is 2

(C) $[\vec{a}\ \vec{b}\ \vec{c}]+[\vec{c}\ \vec{a}\ \vec{b}]=8$

(D) $|3\vec{a}+\vec{b}-2\vec{c}|^2=51$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Which of the following Boolean expressions is not a tautology?

(A) $(p \Rightarrow q) \vee (\sim q \Rightarrow p)$ (B) $(q \Rightarrow p) \vee (\sim q \Rightarrow p)$

(C) $(p \Rightarrow \sim q) \vee (\sim q \Rightarrow p)$ (D) $(\sim p \Rightarrow q) \vee (\sim q \Rightarrow p)$

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

The number of solutions of $\sin^7 x + \cos^7 x = 1$, $x \in [0, 4\pi]$ is equal to

(A) 11 (B) 7 (C) 5 (D) 9

Open solution

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Q86. MCQ (Single Correct Answer) · **Mathematics**

Let a line L : 2x + y = k, k > 0 be a tangent to the hyperbola $x^2 - y^2 = 3$. If L is also a tangent to the parabola $y^2 = \alpha x$, then α is equal to :

(A) 12 (B) -12 (C) 24 (D) -24

Open solution

Q87. Numerical · **Mathematics**

The number of elements in the set $\{n \in \{1, 2, 3, ..., 100\} \mid (11)^n > (10)^n + (9)^n\}$ is _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

Choose the correct option

(A) True dip is always equal to apparent dip.

(B) True dip is not mathematically related to apparent dip.

(C) True dip is less than the apparent dip.

(D) True dip is always greater than the apparent dip.

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

What should be the height of transmitting antenna and the population covered if the television telecast is to cover a radius of 150 km? The average population density around the tower is $2000/km^2$ and the value of $R_e = 6.5 \times 10^6$ m.

(A) Height = 1241 m

Population covered = 7×10^5

(B) Height = 1731 m

Population covered = 1413×10^5

(C) Height = 1800 m

Population covered = 1413×10^8

(D) Height = 1600 m

Population covered = 2×10^5

Open solution

Q90. Numerical · **Physics**

In 5 minutes, a body cools from 75°C to 65°C at room temperature of 25°C. The temperature of body at the end of next 5 minutes is _____° C.

Open solution

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Answer Key

Chemistry

1. **B**	2. **D**	3. **A**	4. **D**	5. **A**
6. **B**	7. **C**	8. **C**	9. **B**	10. **A**
11. **D**	12. **A**	13. **C**	14. **A**	15. **3**
16. **78**	17. **6**	18. **3**	19. **2**	20. **21**
21. **12**	22. **7**	23. **4**		

Mathematics

24. **A**	25. **C**	26. **C**	27. **D**	28. **A**
29. **D**	30. **A**	31. **C**	32. **D**	33. **D**
34. **B**	35. **A**	36. **A**	37. **A**	38. **720**
39. **96**	40. **3125**	41. **4**	42. **1251**	43. **2**
44. **4**	45. **8**	46. **4**		

Physics

47. **B**	48. **C**	49. **C**	50. **C**	51. **C**
52. **A**	53. **D**	54. **D**	55. **C**	56. **C**
57. **B**	58. **A**	59. **A**	60. **D**	61. **A**
62. **B**	63. **C**	64. **D**	65. **60**	66. **1**
67. **3**	68. **500**	69. **4**	70. **2**	71. **15**
72. **05**	73. **02**			

Out of syllabus

74. **D**	75. **A**	76. **A**	77. **B**	78. **B**
79. **A**	80. **106**	81. **B**	82. **A**	83. **D**
84. **D**	85. **C**	86. **D**	87. **96**	88. **C**
89. **B**	90. **57**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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